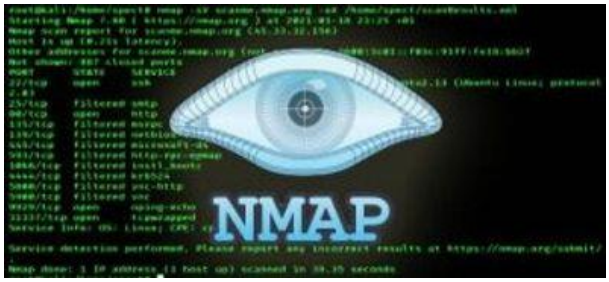
Nmap scan types (-sn, -sS, -sT, -sU, -sV, -O, -A): how each works and what it leaves in packets and logs.

1. Introduction to Nmap

Nmap (Network Mapper) is an open-source tool used for network discovery, port scanning, service identification, and security auditing.



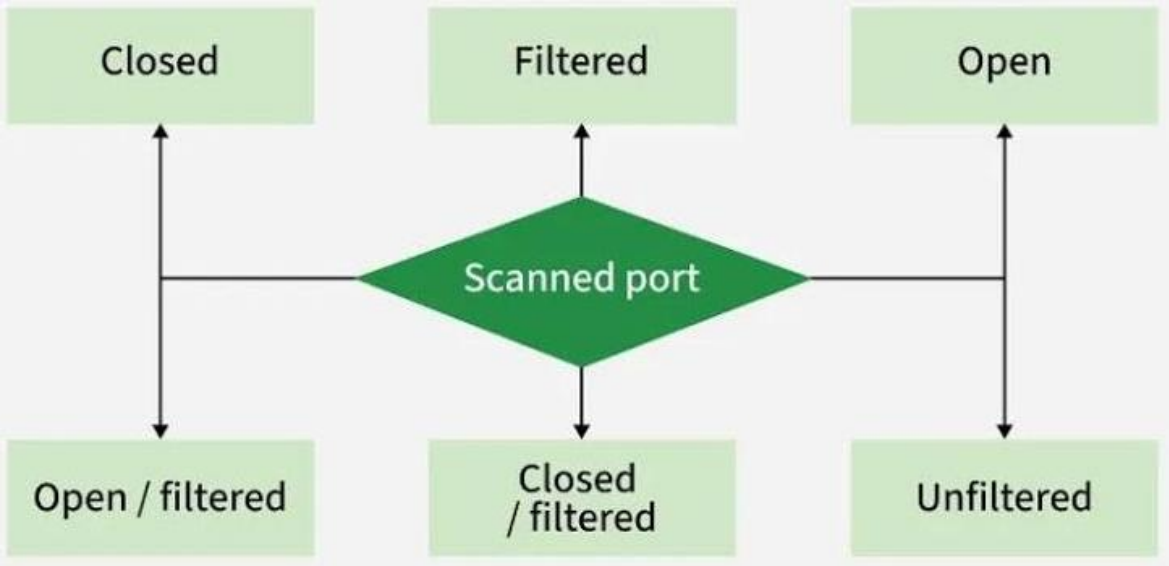
Nmap can help identify:

- Active hosts.
- Open, closed, and filtered ports.
- Running network services.
- Service versions.
- Operating system characteristics.
- Network paths and exposed services.

Nmap scans should be performed only against authorized systems.

2. Nmap Port States

Port State	Meaning
Open	An application is accepting connections or datagrams
Closed	The port is reachable, but no service is listening
Filtered	A firewall or filtering mechanism prevents Nmap from determining the state
Unfiltered	The port is accessible, but its open/closed state is not determined by that scan
Open|Filtered	Nmap cannot distinguish between open and filtered
Closed|Filtered	Nmap cannot distinguish between closed and filtered



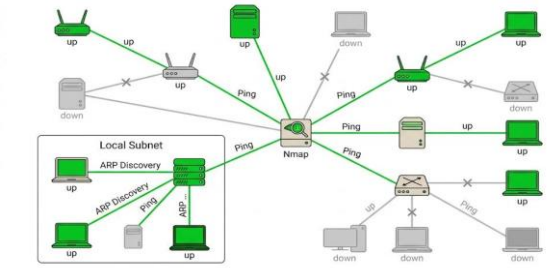
3. Nmap Scan Types

Option	Scan Type	Working Mechanism	Packet/Log Footprint
-sn	Host Discovery	Identifies active hosts without a port scan	ARP, ICMP, or TCP discovery probes
-sS	TCP SYN Scan	Sends SYN probes without normally completing the handshake	SYN, SYN-ACK/RST responses, incomplete connection patterns
-sT	TCP Connect Scan	Establishes TCP connections using the OS network API	Full TCP handshakes on open ports, connection records
-sU	UDP Scan	Sends UDP probes and analyzes responses	UDP traffic, ICMP errors, service responses
-sV	Service Version Detection	Sends protocol-specific probes to identify services	Service banners and application-layer probe traffic
-O	OS Detection	Analyzes TCP/IP fingerprint responses	Characteristic TCP, UDP, and ICMP probes
-A	Aggressive Detection	Enables OS detection, version detection, default NSE scripts, and traceroute	Multiple probes, application requests, and possible IDS alerts

4. Host Discovery (-sn)

nmap -sn 192.168.1.0/24

This scan identifies active devices without scanning their ports.



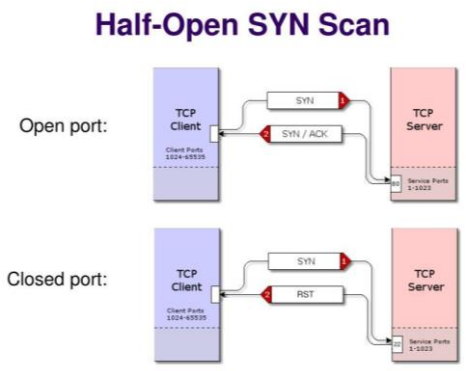
Depending on the network and privileges, Nmap may use ARP, ICMP Echo Requests, or TCP probes.

Security evidence: Repeated discovery packets sent to many hosts may indicate network reconnaissance.

5. TCP SYN Scan (-sS)

sudo nmap -sS 192.168.1.10

A SYN scan sends TCP SYN packets to target ports.



For an open port:

1. Scanner sends SYN.
2. Target responds with SYN-ACK.
3. Scanner normally sends RST instead of completing the handshake.

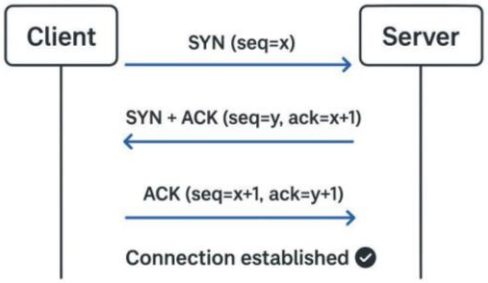
For a closed port, the target generally responds with RST.

Security evidence: Multiple SYN packets directed toward different ports, often without completed connections.

6. TCP Connect Scan (-sT)

nmap -sT 192.168.1.10

This scan uses the operating system's TCP connection functionality.



For open ports, the complete three-way handshake occurs.

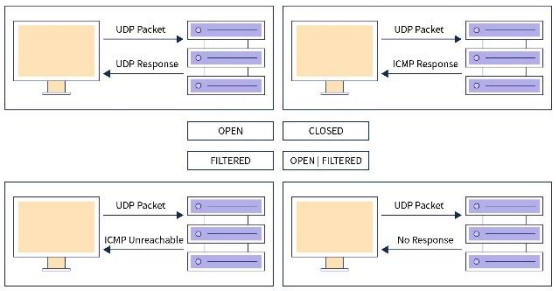
The connection is subsequently closed.

Security evidence: Completed TCP connections, which may be recorded in network and application logs.

7. UDP Scan (-sU)

sudo nmap -sU 192.168.1.10

UDP scanning sends UDP packets to target ports.



Possible results:

- UDP response: Usually indicates an open port.
- ICMP Port Unreachable: Indicates a closed port.
- No response: May indicate an open or filtered port.

UDP scanning can be slower because many UDP services do not respond to unexpected packets.

8. Service Version Detection (-sV)

nmap -sV 192.168.1.10

Nmap sends service-specific probes to identify running applications and versions.

Example:

```
PORT    STATE SERVICE VERSION
22/tcp  open  ssh     OpenSSH 9.2
80/tcp  open  http    Apache httpd 2.4
```

Version detection can generate recognizable application requests and service banners.

9. Operating System Detection (-O)

sudo nmap -O 192.168.1.10

Nmap estimates the target operating system by analyzing network-stack behavior.

It examines characteristics such as:

- TCP window sizes.
- TCP options.
- Initial TTL values.
- Responses to unusual probes.
- IP identification behavior.

Results are estimates and may not always be accurate.

10. Aggressive Detection (-A)

sudo nmap -A 192.168.1.10

Aggressive detection enables:

- Operating system detection.
- Service version detection.
- Default NSE scripts.
- Traceroute.

It may produce more traffic and log entries than basic scanning.

11. Nmap Evidence in Network Logs

Nmap scanning may leave:

- Firewall connection logs.
- IDS/IPS alerts.
- TCP SYN packets across multiple ports.
- ICMP discovery traffic.
- UDP probes and ICMP errors.
- Application requests generated by version detection.
- Unusual connection attempts across many hosts.

The exact footprint depends on scan options, privileges, target behavior, firewalls, and monitoring configuration.